Tribhuvan University

Faculty of Humanities and Social Sciences

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A Lab Report

On

Programming Logic & Techniques using Python (MCA504)

Submitted by

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Assignment 1

1. Assume we have 10 US dollars & 25 Saudi Riyals. Write a Python Script to convert the total currency to Nepali Rupees. (Exchange rates: 1 US Dollar = 133.72 NRs and 1 Saudi Riyal = 35.82 NRs)

```
dollar = 10
riyal = 25
nepali_rupees = (dollar * 133.72) + (riyal * 35.82)
print(f"Total currency in NRs: {nepali_rupees}")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Total currency in NRs: 2232.7
```

2. Given the initial deposit, the annual interest rate, and the number of years that you invest your money in an investment account. Calculate the final amount of the investment using math.pow() function.

```
import math
deposit = float(input("Enter the initial deposit: "))
interest_rate = float(input("Enter the annual interest rate (in decimal): "))
years = int(input("Enter the number of years: "))

final_amount = deposit * math.pow((1+ interest_rate), years)
print(f"The final amount of the investment after {years} years is: {final_amount:.2f}")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter the initial deposit: 100000
Enter the annual interest rate (in decimal): 3
Enter the number of years: 2
The final amount of the investment after 2 years is: 1600000.00
```

3. Write a program to find root of the equation 4x**2 -21x + 25=0 using the formula x = (-b ± √(b² - 4ac)) / 2a

```
import math
a = 4
b = -21
c = 25

discriminant = b**2 - 4*a*c
if discriminant > 0:
    root1 = (-b + math.sqrt(discriminant)) / (2*a)
    root2 = (-b - math.sqrt(discriminant)) / (2*a)
    print(f"The roots are real and different: {root1} and {root2}")
elif discriminant == 0:
    root = -b / (2*a)
    print(f"The roots are real and the same: {root}")
else:
    real_part = -b / (2*a)
    imaginary_part = math.sqrt(-discriminant) / (2*a)
    print(f"The roots are complex: {real_part} + {imaginary_part}i and {real_part} - {imaginary_part}i")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
The roots are real and different: 3.425390529679106 and 1.824609470320894
```

4. Write a program to take "gender" as input from user. If the user is male, give the message: Good Morning Sir. If the user is female, give the message: Good Morning Ma'am.

```
gender = input("Enter your gender: ")

if gender == "male":
    print("Good Morning Sir.")
elif gender == "female":
    print("Good Morning Ma'am.")
else:
    print("Invalid input. Please enter only male or female.")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter your gender: male
Good Morning Sir.

C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter your gender: female
Good Morning Ma'am.
```

5. Write a program to take input color of road traffic signal from the user & show the message according to this table:

SIGNAL COLOR	MESSAGE
RED	Vehicle must stop
YELLOW	Vehicles should get ready to move
GREEN	Vehicles can move now

```
signal_color = input("Enter the color of the traffic signal (RED, YELLOW, GREEN): ").upper()
if signal_color == "RED":
    print("Vehicle must stop")
elif signal_color == "YELLOW":
    print("Vehicles should get ready to move")
elif signal_color == "GREEN":
    print("Vehicles can move now")
else:
    print("Invalid input. Please enter RED, YELLOW, or GREEN.")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter the color of the traffic signal (RED, YELLOW, GREEN): red
Vehicle must stop

C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter the color of the traffic signal (RED, YELLOW, GREEN): green
Vehicles can move now

C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter the color of the traffic signal (RED, YELLOW, GREEN): yellow
Vehicles should get ready to move
```

6. Write a program that takes user input day name. If the day is Monday, Tuesday, Wednesday, Thursday or Friday, then show “It’s a week day”. If the day is Saturday then show “It’s weekend”. If the day is sunday then show “Yay! It’s a holiday”.

```
day = input("Enter the day name: ").upper()
days = ["MONDAY", "TUESDAY", "WEDNESDAY", "THURSDAY", "FRIDAY"]
if day in days:
   print("It's a week day")
elif day == "SATURDAY":
   print("It's weekend")
elif day == "SUNDAY":
   print("Yay! It's a holiday")
else:
   print("Invalid day name")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>p
Enter the day name: sunday
Yay! It's a holiday

C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>p
Enter the day name: monday
It's a week day

C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>p
Enter the day name: saturday
It's weekend
```

7. Use a conditional (ternary) operator for this exercise:

If the variable age is a value below 18, the value of the variable voteable should be "Too young", otherwise the value of voteable should be "Old enough".

```
age = int(input("Enter your age: "))

votable = "Too young" if age < 18 else "Old enough"
print(votable)
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter your age: 15
Too young

C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter your age: 19
Old enough
```

8. Write a program to take input the marks obtained in three subjects & total marks. Compute & show the resulting percentage on your page. Take percentage & compute grade as per following table:

Percentage	Grade	Remarks
Greater than or equal to 80	A one	Excellent
Greater than or equal to70	A	Good
Greater than or equal to 60	B	You need to improve
Less than 60	Fail	Sorry

```
marks1 = float(input("Enter marks obtained in subject 1: "))
marks2 = float(input("Enter marks obtained in subject 2: "))
marks3 = float(input("Enter marks obtained in subject 3: "))

total_marks = float(input("Enter total marks: "))

total_obtained = marks1 + marks2 + marks3
percentage = (total_obtained / total_marks) * 100
print(f"Total Marks Obtained: {total_obtained}")
print(f"Percentage: {percentage:.2f}%")

if percentage >= 80:
    print("Grade: A-one")
    print("Remarks: Excellent")
elif percentage >= 70:
    print("Grade: A")
    print("Remarks: Good")
elif percentage >= 60:
    print("Grade: B")
    print("Remarks: You need to improve")
else:
    print("Grade: Fail")
    print("Remarks: Sorry")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter marks obtained in subject 1: 67
Enter marks obtained in subject 2: 89
Enter marks obtained in subject 3: 90
Enter total marks: 300
Total Marks Obtained: 246.0
Percentage: 82.00%
Grade: A-one
Remarks: Excellent
```

9. Write a program to repeatedly print the value of the variable num which is input by user. Value should be decreasing by 0.5 each time, as long as x Value remains positive.

```
num = float(input("Enter a positive number: "))
while num > 0:
    print(num)
    num -= 0.5
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
Enter a positive number: 5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
```

10. Write a program using for loop that will iterate from 1 to 20. For each iteration, it will check if the current number is even or odd, and report that to the screen (e.g."2 is even").

1 is Odd
2 is Even
3 is Odd
4 is Even
5 is Odd
and so on.

```
for i in range(1, 21):
    if i % 2 == 0:
        print(f"{i} is Even")
    else:
        print(f"{i} is Odd")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 1>
1 is Odd
2 is Even
3 is Odd
4 is Even
5 is Odd
6 is Even
7 is Odd
8 is Even
9 is Odd
10 is Even
11 is Odd
12 is Even
13 is Odd
14 is Even
15 is Odd
16 is Even
17 is Odd
18 is Even
19 is Odd
20 is Even
```